

Day 19 - Lecture 19.10: Introduction to Seaborn

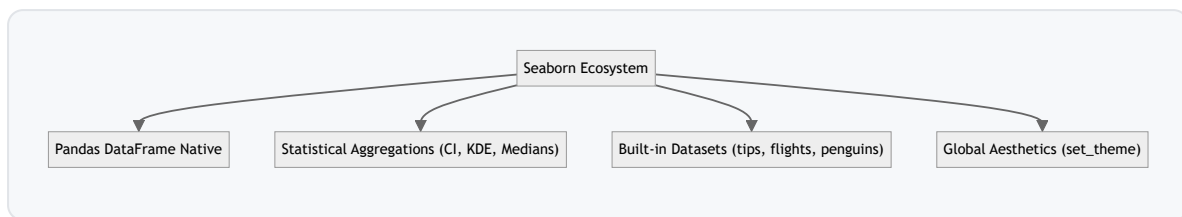
1. Overview & What is Seaborn?

Seaborn is a Python data visualization library based on **Matplotlib**. It provides a high-level interface for drawing attractive and informative statistical graphics.

Matplotlib vs. Seaborn:

Feature	Matplotlib	Seaborn
Level	Low-level (Direct control over pixels, lines, axes)	High-level (Abstracted statistical functions)
Data Format	NumPy arrays, Python lists, series	Native Pandas DataFrame support (column names as strings)
Statistical Computations	Manual (user must calculate mean, error bars, regression)	Built-in (calculates confidence intervals, aggregation, KDE)
Styling	Plain default styling, requires manual configuration	Modern, aesthetic themes out-of-the-box
Multi-dimensional Data	Requires manual loops for colors/markers	Handled effortlessly with <code>hue</code> , <code>style</code> , <code>size</code> , <code>col</code>

2. Core Features & Themes



1. Setting Themes (`sns.set_theme()`):

- **Styles:** `darkgrid` (default), `whitegrid`, `dark`, `white`, `ticks`
- **Palettes:** `deep`, `muted`, `bright`, `pastel`, `dark`, `colorblind`
- **Context:** `paper`, `notebook` (default), `talk`, `poster` (scales fonts for presentations)

2. Inbuilt Datasets:

Seaborn includes repository datasets accessible via `sns.load_dataset(name)` :

- `tips` : Waiter tips data (restaurant bills, tips, sex, smoker status, day, time).
 - `flights` : Monthly airline passengers from 1949 to 1960.
 - `penguins` : Palmer Archipelago penguin measurements (flipper length, bill length, body mass).
 - `iris` : Classic Fisher Iris flower measurements.
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3. Code Implementation

```
import seaborn as sns
import matplotlib.pyplot as plt

# 1. Check version and available datasets
print("Seaborn Version:", sns.__version__)
print("First 10 datasets:", sns.get_dataset_names()[:10])

# 2. Setting modern aesthetic theme
sns.set_theme(style="whitegrid", palette="muted")

# 3. Loading the tips dataset
tips = sns.load_dataset("tips")
print("Dataset Info:")
print(tips.head())
print(tips.info())

# 4. Quick introductory plot
plt.figure(figsize=(8, 5))
sns.scatterplot(data=tips, x="total_bill", y="tip", hue="time", style="time")
plt.title("Tips: Total Bill vs Tip Amount by Dining Time", fontsize=14, fontweight="bold")
plt.show()
```

4. Key Takeaways

- Seaborn does not replace Matplotlib—it **extends** it. Every Seaborn plot is drawn on Matplotlib `Figure` and `Axes` objects.
- Always use `sns.set_theme()` at the start of your notebooks to instantly upgrade plot aesthetics.